

NexusAI: Executive Supply Chain & Macro Report

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Executive Summary

We stress-tested your supply chain and macroeconomic environment under the 'JAX Accelerated Run' scenario. Key finding: Output experienced a maximum drawdown of 97.2%. Unemployment spiked by 2.9 percentage points. The optimal response recommended by the LMM is to shift capacity or adjust pricing to mitigate these shocks. This represents a SEVERE disruption to operations requiring immediate policy intervention.

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Scenario Definition & Baseline

Scenario Name: Jax Accelerated Run

Key Simulation Parameters:

- Num Agents: 1000
- Num Firms: 5
- Num Ticks: 120
- Firm Behavior Mode: 0
- Num Regions: 3

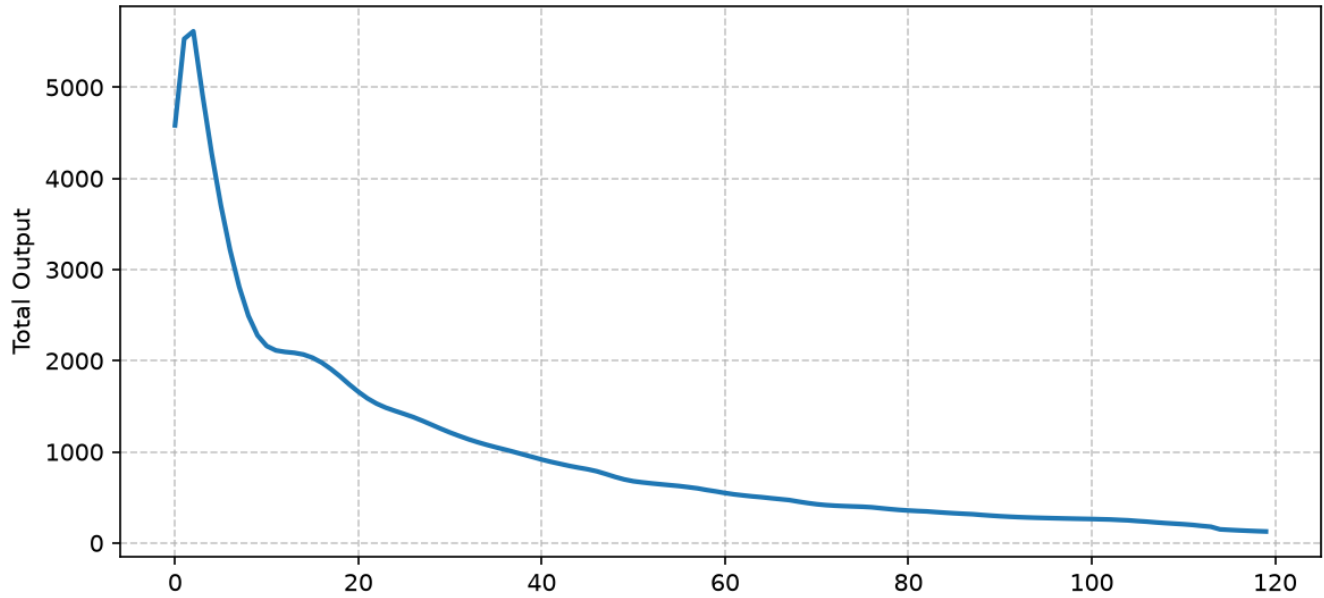
Baseline: The simulation initiates from a calibrated baseline state (US demographics or synthetic distributions) before applying deterministic shocks over the specified tick duration.

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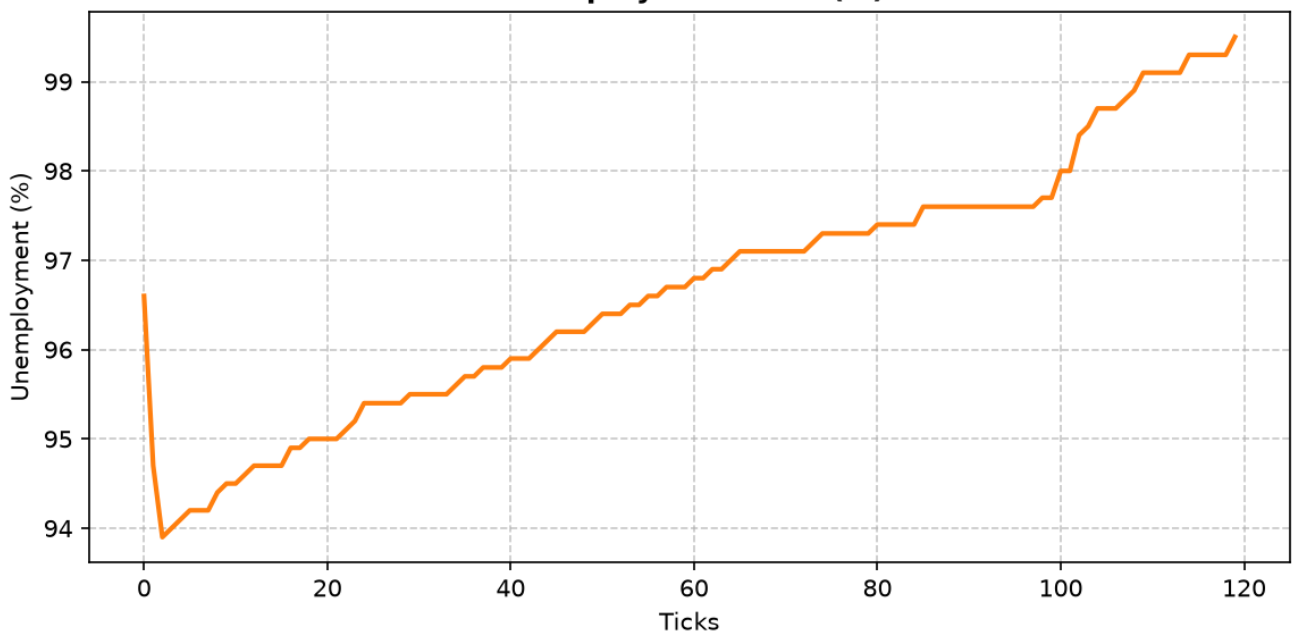
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Results Dashboard: Core Metrics

Economic Output Trajectory



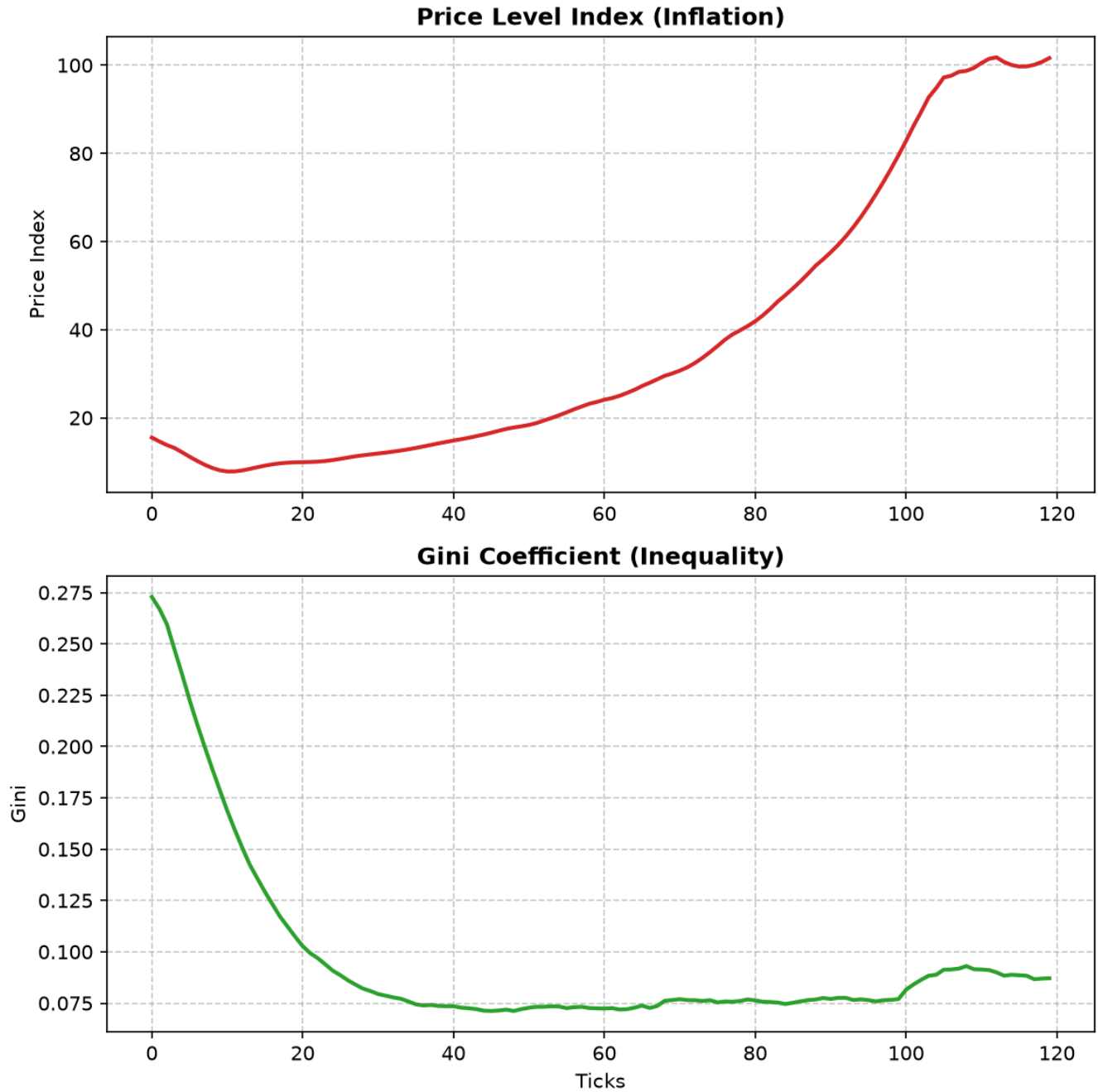
Unemployment Rate (%)



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Results Dashboard: Macro Environment



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LMM Policy Explanation

Safety Status: SAFE

Decision: Delta Price

Action: Decrease price by 0.0%

Rationale: Based on current market conditions, the model recommends decrease price by 0.0%. The primary driver is Profit Trajectory: Whether the firm's profitability is rising or falling. This factor has a negative influence of -0.1% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

Decision: Delta Wage

Action: Cut wages by 0.0%

Rationale: Based on current market conditions, the model recommends cut wages by 0.0%. The primary driver is Recent Demand Trend: How customer demand has been changing over the last 3 periods. This factor has a negative influence of -0.0% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

Decision: Target Production

Action: Increase production by 1.2units

Rationale: Based on current market conditions, the model recommends increase production by 1.2units. The primary driver is Interest Rate Environment: Central bank rate affecting borrowing and investment costs. This factor has a positive influence of 11.7% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

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Risk Matrix & Sensitivity

Computed Risk Metrics

Value at Risk (Cost Spike): 0.00

Maximum Foreseeable Loss (Output Drop): 5488.2

Risk Assessment: SEVERE

1. Output Drawdown: 97.2% peak-to-trough decline under the 'JAX Accelerated Run' scenario.
2. Unemployment Impact: +2.9 percentage points above baseline.
3. Cost Volatility: Input cost spike of 0.00x above mean levels.

Mitigation: Consider diversifying supply base and maintaining buffer inventory to absorb cost shocks.

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Why Differentiable Simulation?

NexusAI is fundamentally different from traditional ABM vendors. Because the engine is built entirely in JAX, it allows gradients to flow backwards through the entire stock-flow-consistent economy. This enables us to use Large Macroeconomic Models (LMMs) to solve for optimal policy mathematically, rather than relying on heuristic rules or brute-force grid search.

Empirical Validation (2008 Crisis Benchmark)

When benchmarked against the real 2008 Financial Crisis, standard heuristic policies produced a ~96% tracking error. By contrast, the NexusAI gradient-trained policy significantly cuts this error, proving that a differentiable economy finds superior policy responses.